Adroit Technologies Protocol Driver

Name	DNP3 Level 2+
DLL	DNP3

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Contents

1. Overview

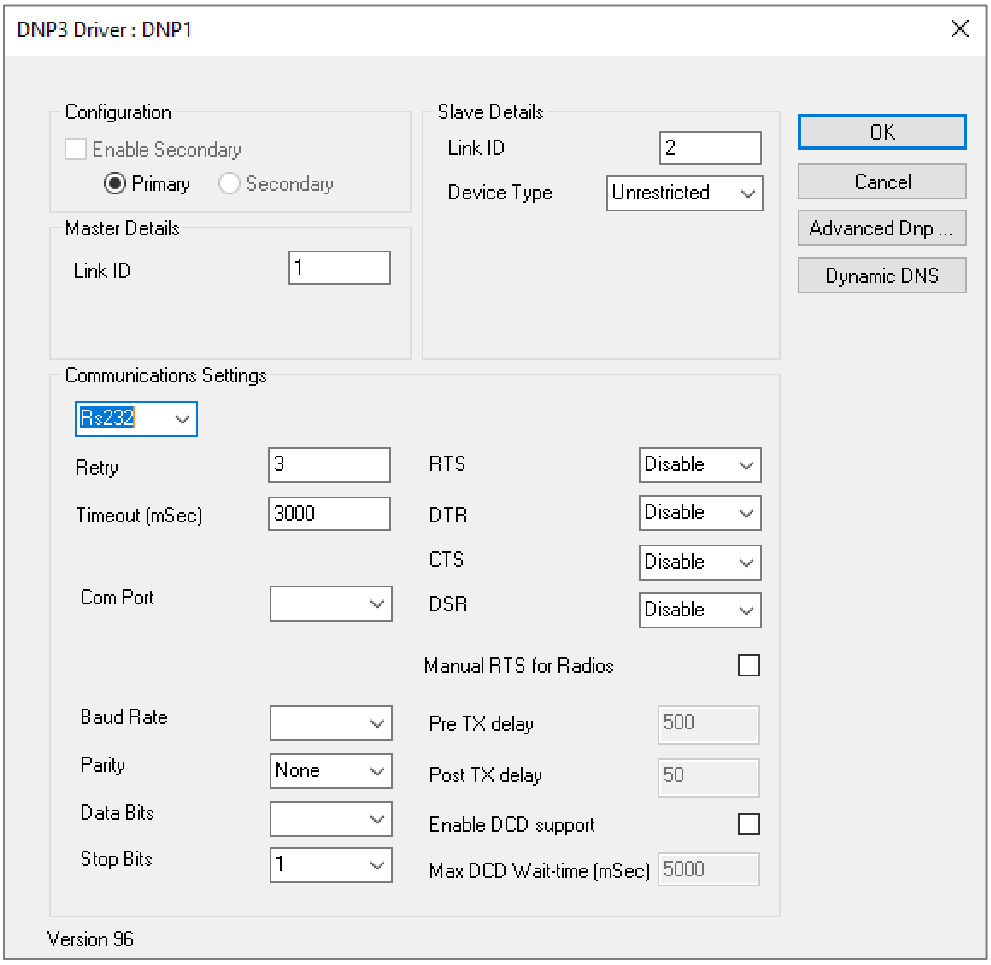
Adroit supports communication with the following DNP3 levels : Level 1 and level 2 and some level 3 and 4 functions.

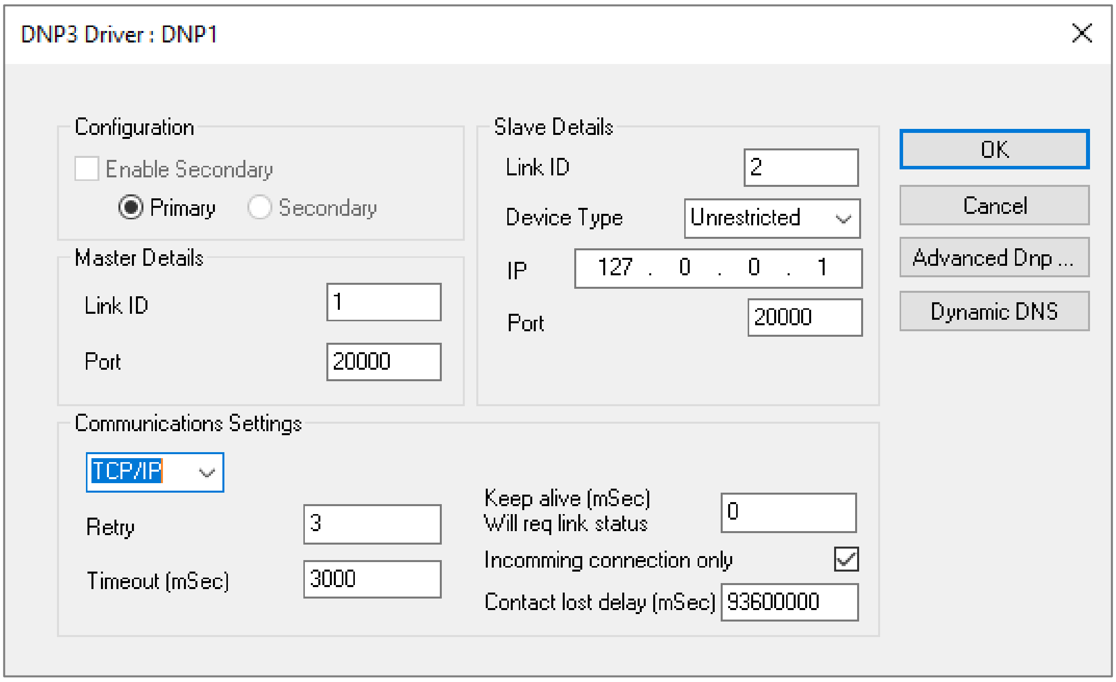
2. Wiring

Standard depending on configuration. DNP3 can be utilised with RS-485, RS-232, Modem (RS-232), and Ethernet.

3. Device Configuration

3.1. Configuration





Enable Secondary – Will allow configuration for a second communication resource that will allow for dual redundancy at the driver level. The device will then be able to switching between configurations when a completed retry cycle exits without response from the slave device.

Primary / Secondary – Selects between the two configurations (Enable Secondary must be enabled.)

3.2. Master Details

Link ID - Master identification number.

Port – (Ethernet only) The TCP or UDP port number to use on the master side.

Com Type – Choice between Rs232, Rs485, Modem(Rs232) or Ethernet communications resource.

3.3. Slave Details

Link ID - Slave identification number.

Device Type - Unrestricted currently implements protocol maximums device specific maximums and default will become available when available.

IP - (Ethernet only) Remote IP address.

Port - (Ethernet only) Only available ports will be listed here.

3.4. Communication Settings

Communication Combo box: RS232, RS485, TCP/IP, UDP/IP, MODEM.
Some group box areas will dynamically update to suit the communications resource selected.

Retry - The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Timeout - The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write). This is referring to time allowed to receive individual transport fragments that make up and application fragment. *See **Application timeout** in the advanced dialogue for the overall Application timeout.*

3.4.1. RS-232, RS-485, MODEM Specific Parameters:

Com Port – Select as per requirement.

Baud Rate – Select as per requirement.

Parity – Select as per requirement.

Databits – Select as per requirement.

Stopbits – Select as per requirement.

RTS – If enabled, the RTS signal line will be permanently on. If disabled, the RTS signal line will be permanently off. Other options are Toggle or Handshake as per requirement.
Manual RTS for Radios – (Rs232 only) RTS will be asserted according to the delays specified below. This will allow the RTS to signal the radio key.

Pre TX delay – (Rs232 only) (only if Manual RTS for Radios is checked) Key up time before manual RTS is Asserted (Read or Write).

Post TX delay - (Rs232 only) (only if Manual RTS for Radios is checked) Key down time before manual RTS is de-asserted after transmission (Read or Write).

DTR – (Serial types only) – If enabled, the DTR signal line will be permanently on. If disabled, the DTR signal line will be permanently off. Other option is Handshake as per requirement.

CTS – (Serial types only) If enabled, the CTS signal line will be permanently on. If disabled, the CTS signal line will be permanently off.

DSR – If enabled, the DSR signal line will be permanently on. If disabled, the DSR signal line will be permanently off.

Rs485 Reflections – (Rs485 only) If checked, reflections are expected and will be filtered out during on any read algorithm. Currently no reflection checking is enabled.

Enable DCD support – (Default = OFF) If checked, it will enable data carrier detection so that the driver attempts to prevent clashes (especially when using radios).

Max DCD Wait Time (mSec) – Maximum backoff time to wait while a carrier busy is detected. If this time is exceeded before transmission, the transaction will fail, and normal retry procedures will be followed.

3.4.2. TCP/IP, UDP/IP Specific Parameters:

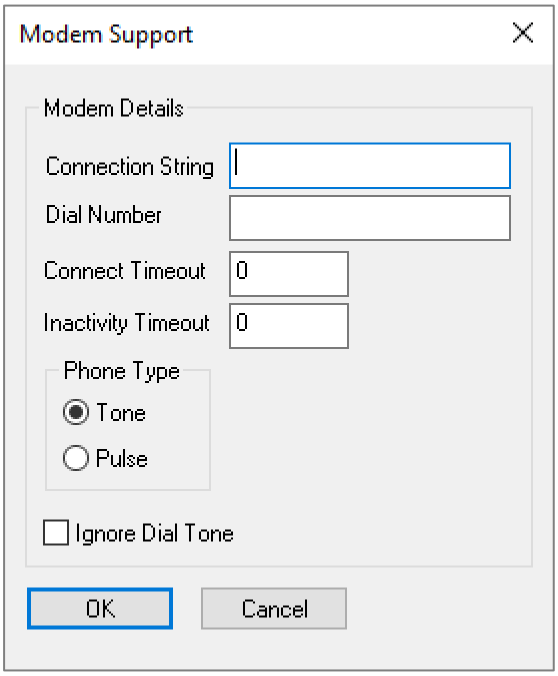
Keep Alive – (Ethernet types only) The period (in milliseconds) the driver will wait between keepalive (LINK TEST) messages.

Incoming connection only – (Ethernet types only)If checked the driver will wait for incoming connections, it will not initiate connections, used for internet of things type devices that connect to the master only periodically, see Contact lost setting too. The driver will operate as normal when a connection is established, if the outstation disconnects the driver will wait for the next connection to continue. The master will not disconnect, that function is left up to the slave (Default = unchecked).

Contact lost delay (mSec) – (Ethernet types only) Only active if the Incoming connection only is checkbox is checked. This is the period (in milliseconds) the driver will wait before marking I/O point as comms fail (Default = 93600000 = 26 hours).

<u>*If Modem communication resource was selected:*</u>

Modem Button – When clicked you will be able to setup the appropriate modem configuration details. See the Modem Support dialog menu below:



3.5. Modem Details

Connection string – The string used in configuring the modem on each dialup attempt. This will allow modem specific configurations before the dial attempt take place.

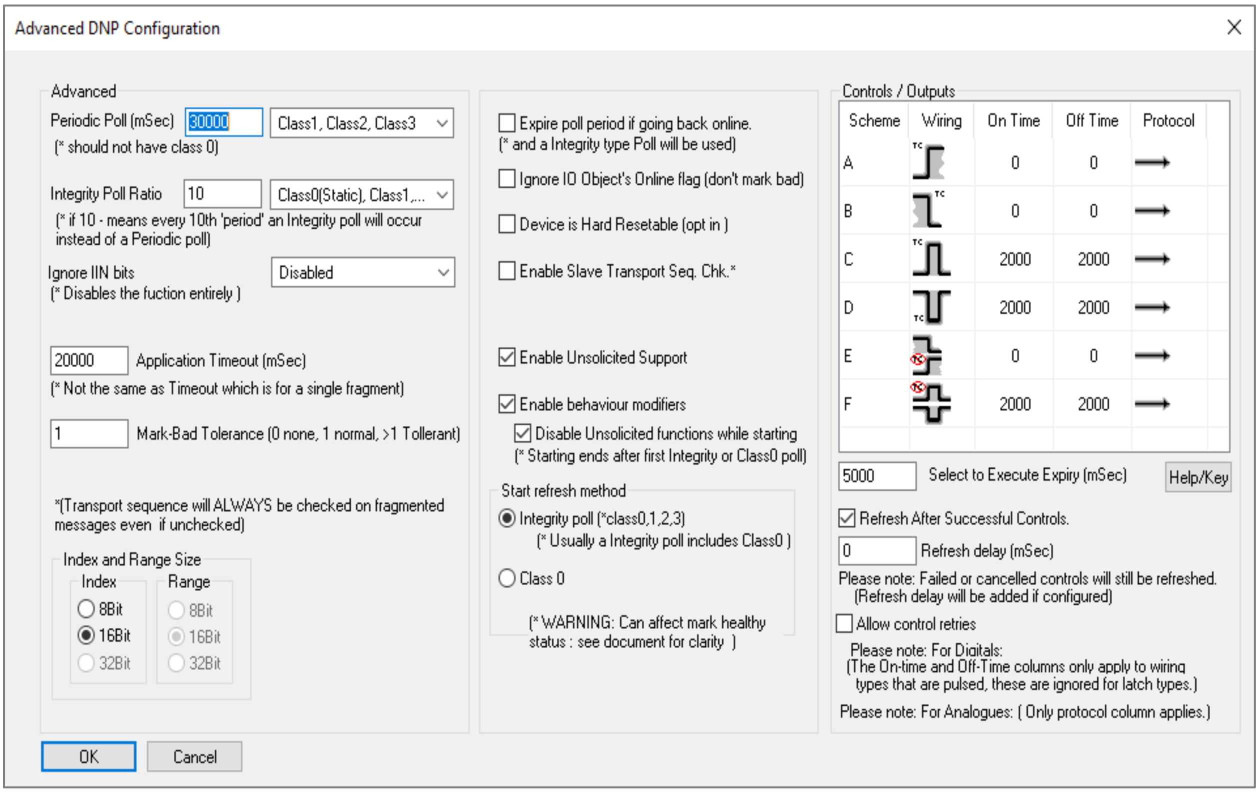
Dial Number – The destination telephone number.

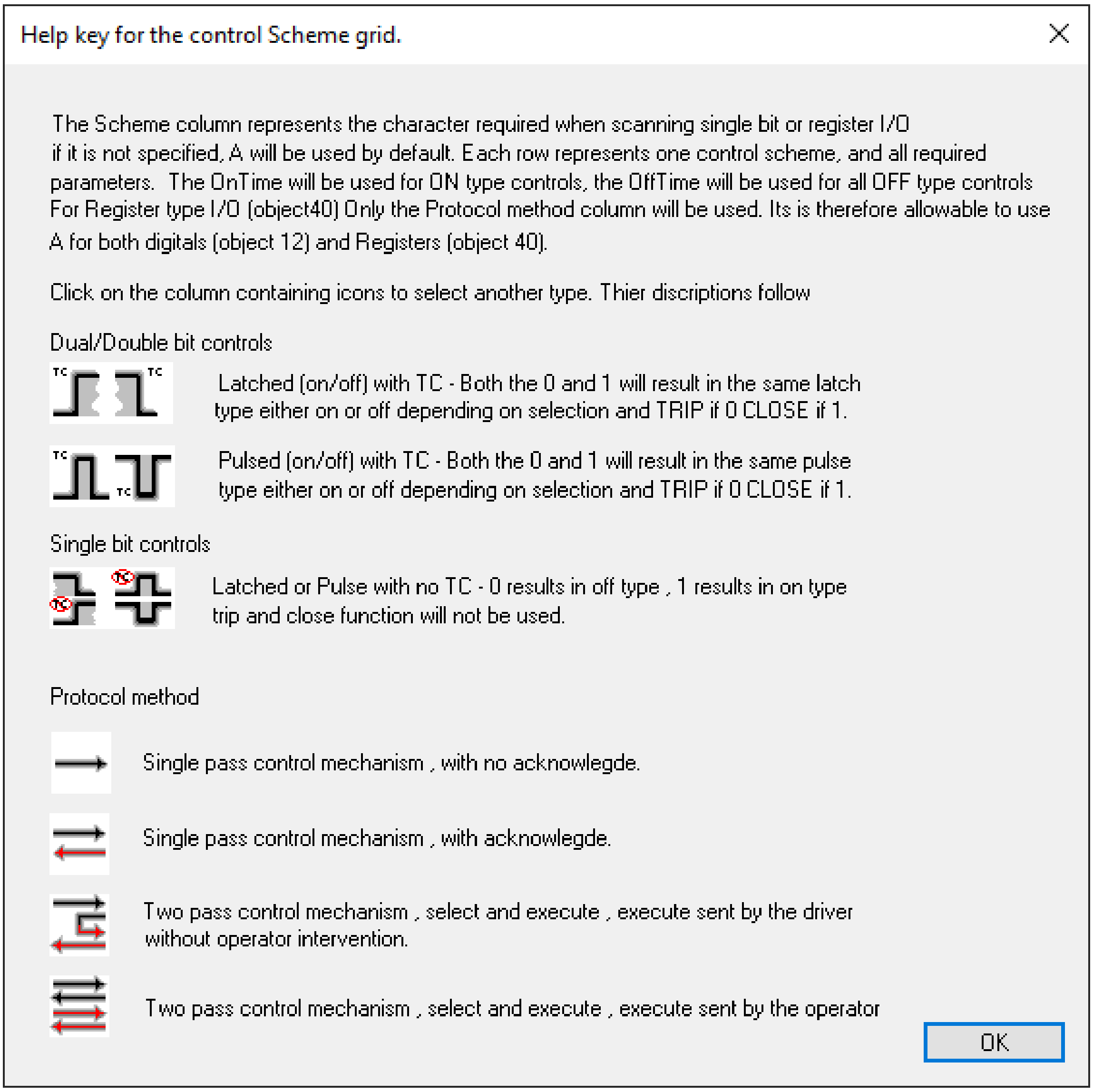
Connection Timeout – (milliseconds) The maximum period allowed to dial.

Inactivity Timeout – (milliseconds) The inactivity period before the modem auto hangs up.

Phone Type – Tone or Pulse mode.

Ignore Dial Tone – Set if dial tone is unavailable or is at an unreliable level.


Advanced DNP Configuration
Advanced
Periodic Poll (mSec) 30000 Class1, Class2, Class3
(* should not have class 0)
Integrity Poll Ratio 10 Class0(Static), Class1,...
(* if 10 - means every 10th 'period' an Integrity poll will occur instead of a Periodic poll)
Ignore IIN bits Disabled
(* Disables the fuction entirely)
20000 Application Timeout (mSec)
(* Not the same as Timeout which is for a single fragment)
1 Mark-Bad Tolerance (0 none, 1 normal, >1 Tollerant)
*(Transport sequence will ALWAYS be checked on fragmented messages even if unchecked)
Index and Range Size
Index 8Bit 16Bit 32Bit
Range 8Bit 16Bit 32Bit
Expire poll period if going back online.
(* and a Integrity type Poll will be used)
Ignore IO Object's Online flag (don't mark bad)
Device is Hard Resetable (opt in)
Enable Slave Transport Seq. Chk.*
Enable Unsolicited Support
Enable behaviour modifiers
Disable Unsolicited functions while starting
(* Starting ends after first Integrity or Class0 poll)
Start refresh method
Integrity poll (*class0,1,2,3)
(* Usually a Integrity poll includes Class0)
Class 0
(* WARNING: Can affect mark healthy status : see document for clarity)
Controls / Outputs
Scheme Wiring On Time Off Time Protocol
A 0 0
B 0 0
C 2000 2000
D 2000 2000
E 0 0
F 2000 2000
5000 Select to Execute Expiry (mSec) Help/Key
Refresh After Successful Controls.
0 Refresh delay (mSec)
Please note: Failed or cancelled controls will still be refreshed. (Refresh delay will be added if configured)
Allow control retries
Please note: For Digitals: (The On-time and Off-Time columns only apply to wiring types that are pulsed, these are ignored for latch types.)
Please note: For Analogues: (Only protocol column applies.)
OK Cancel


Help key for the control Scheme grid.
The Scheme column represents the character required when scanning single bit or register I/O if it is not specified, A will be used by default. Each row represents one control scheme, and all required parameters. The OnTime will be used for ON type controls, the OffTime will be used for all OFF type controls For Register type I/O (object40) Only the Protocol method column will be used. Its is therefore allowable to use A for both digitals (object 12) and Registers (object 40).
Click on the column containing icons to select another type. Thier discriptions follow
Dual/Double bit controls
Latched (on/off) with TC - Both the 0 and 1 will result in the same latch type either on or off depending on selection and TRIP if 0 CLOSE if 1.
Pulsed (on/off) with TC - Both the 0 and 1 will result in the same pulse type either on or off depending on selection and TRIP if 0 CLOSE if 1.
Single bit controls
Latched or Pulse with no TC - 0 results in off type , 1 results in on type trip and close function will not be used.
Protocol method
Single pass control mechanism , with no acknowlegde.
Single pass control mechanism , with acknowlegde.
Two pass control mechanism , select and execute , execute sent by the driver without operator intervention.
Two pass control mechanism , select and execute , execute sent by the operator
OK

3.6. Advanced

Periodic Poll (mSec) - Insert the required period in milliseconds here to cause the driver to initiate a periodic poll which consists of any (or all) classes defined in **the dropdown check combo** alongside.

Integrity Poll Ratio - Insert the required ratio of periodic to integrity polls which consists of any or all classes defined in the **dropdown check combo** alongside. E.g. if 10 is used, every 10^{th} periodic poll will be an Integrity poll and not periodic. (Note: integrity poll should have class 0 data configured)

Ignore IIN bits - dropdown check combo – select the required bits to be ignored when reported via the IIN flags in the protocol. The appropriate admin bit(s) will have to monitored manually within Adroit, and class bits will be retrieved manually or via script or schedule etc.

> **WARNING**
>
> If selected it is possible the number of events will overflow inside the PLC/RTU buffers, causing a possible loss of data. This will have to be managed carefully.

> **WARNING**
>
> If Timesync is ignored, this is globally ignored, and no Timesync will ever be sent. Use with caution as some objects get time at source and some do not, so mismatched/sync issues become apparent as a result.

Application Timeout (mSec) - (Default = 20000 mSec) The time the driver will allow to receive ALL application fragments of any specific message. Even though application fragments are atomic, the driver will consume them as such but if the last fragment exceeds this timeout it will likely be rejected (Default = 20000 mSec).

Markbad Tolerance - (Default = 1) 0 means no bad marking at all. Users should then refer to the online bit in admin.0 to display comms fails on a mimic and not device.commsfail. 1 means normal bad marking on every fail, while greater than 1 means there will be a tolerance applied to bad marking. Consecutive fails are counted, and the device will have all jobs marked bad when this count reaches the tolerance value.

Index and Range size - Select the types required. Note Level 2 dictates that if 8-bit addressing is used, 8-bit registers must be selected (hence the driver auto selects the correct corresponding range). The driver however will be able to interpret any combination.

Expire poll period if going back online - (Default = unchecked) If checked, it will force the poll period to expire so that a poll is immediately requested to refresh data, and the integrity poll is sent if configured. Note: integrity poll should have class 0 data configured).

Device Hard reset-able - if checked, it will allow hard reset functionality. Note: some devices do reset safely and require manual intervention if reset, hence this function is configurable.

Enable Slave Transport Sequence Check - If checked, it will force the master to check and increment transport sequence on all data received/sent, but it will not reject new application messages unless the failure occurred within a transport fragmentation session. In other words, The master will always strictly enforce transport sequence within a fragmented session.

Enable unsolicited support – (Default = unchecked) If checked, the driver will consume unsolicited/report by exception messages from the outstation.

Enable behaviour modifiers – (Default = checked) If unchecked/opt-out, stops additional functions such as:

Disable unsolicited functions while starting – If unchecked/opt-out, it stops the driver from forcing unsolicited reporting off (but is on during start-up).

Start refresh method : (Default = Integrity poll method checked)

Integrity poll method: Driver will send an integrity poll when starting up, going back online or starting the device from the configurator. This is usually Class 0, Class 1, Class 2 and Class 3 combined in one packet. *See Integrity Poll combo box for message content.*

Class 0 *(ONLY)* method: driver will send a Class 0 poll when starting up, going back online, or starting the device from the configurator. This is used to distinguish between Integrity and Class 0 where the latter requests current status of ALL I/O without time. This can be used for system where the timestamp is important and where sending a ratio of integrity periodic polls might add extra events to the logs, etc,, where one can then separate it. Note that if this is the case, then remove **Class 0** from the **Integrity poll combo**.

⚠	**WARNING**
If all data should be allocated to another Class 1, 2, 3 and the appropriate deadband is applied, then note that if this is used, there may be a period where I/O will look static due to the absence of "integrity-class0" objects in the periodic polls.	

3.7. Controls and Outputs

3.7.1. Scheme Table:

Each row consists of 5 columns: **Scheme, Wiring, OnTime, OffTime** and **Protocol.**

Each row represents 1 scheme. There are 6 available for use with digitals and analogue outputs, and the same scheme can be used for both digital and analogue. However, for analogue controls, only the protocol column has any useful effect, and the other columns are ignored.

⚠	**WARNING**

Multiple Tags scanned to the same DNP address will override the schema for all duplicated tags, and currently it is the schema of last tag to be scanned wins. The application event log will have a warning entry describing the duplication, as will the datascope output.

Scheme - A through F – scanning controls must specify this scheme character in the scanning address, i.e. "40.2.1 A" – Analogue output on control scheme A, if Fig3 were to use it, it would mean Direct operate with acknowledge. (See Fig4 for detailed descriptions).

Wiring – Click on the icons to toggle mode. Defines what type of digital waveform is required, i.e. Pulsed or Latched. It also defines type for both digital states. Dual bit controls will employ trip close as well. (see Fig4).

OnTime – Edit to change. Defines what the duration of the pulse ON for the digital waveform will be. If pulse ON only selected for wiring, then OffTime will be ignored.

OffTime – Edit to change. Defines what the duration of the pulse OFF for the digital waveform will be. If pulse OFF only selected for wiring, then OnTime will be ignored.

Protocol – Click on icon to toggle the mode. Defines the type of protocol mechanism to use Direct/Select & Execute, etc. Register type controls to object 40 will only use this column. The others will be ignored.
Select & Execute - If selected in a scheme, controls will have a select phase and then either an internal or external execute phase.

Driver will execute (internal) - If selected, the driver will confirm the correct control internally. It will then send a corresponding execute without user intervention, else if **external** is selected, then the execute requires a separate user initiated confirmation via the UI where the user has a choice to execute or abort the control. This functionality will require the use of specific bits within the administrative control words.

SelectedBit will be set by the driver when the control has been selected on the first phase, then a visibility should be enabled (on **SelectedBit**) for the common execute button, which in turn sets the common **ExecuteBit**.The driver is waiting for this bit to complete the second and final execute phase, where upon completion the driver will clear both bits and **ControlFailedBit** if it were set from a previous control attempt. If the control fails or times out, the driver will reset both SelectedBit and ExecuteBit, as well as set the ControlFailed Bit to true. All of these bits should provide the required and adequate information back to the user.
Select to execute expiry - (mSec) is the maximum delay between a select and an execute. If this delay expires, the SelectedBit will be reset and the control will be failed, and the ControlFailedBit will be set.

Refresh after successful controls - The associated output status object will be polled (Default = checked = ON).

Refresh Delay (mSec) - Means immediately after a control occurs the refresh will take place, else the refresh will be delayed by the specified amount of time (Default = 0 mSec).

Allow control retries -If checked, then retries will be used. Be very careful with this, as it is not recommended practice to retry controls in DNP3 (Default = unchecked = no retries).

3.8. Maximum Addresses

PLC Type	Max Digitals Per Scan	Max Registers Per Scan)
Unrestricted	Config dependent	Config dependent

3.9. Unique Driver Specifications

1. This driver has extra information and special functions that reside in a special fixed administration integer job, which has address ADMIN.0, ADMIN.1

   These Admin words can be described as control words and are only scan-able into Integer type slots. Marshal agents are recommended so that each bit-slots allows access to each bit individually. These words can force the driver to do additional tasks such as poll for a specific class of data or even request a reset of device. They also provide additional information such as an online bit, which is set whenever a valid interrogation or unsolicited message is consumed. Also it provides the functionality to enable the external select execute operations.

   - **Word 0 bit : [ADMIN.0]**
     1. DeviceOnline (Read only Status)
        a. [1 : Online , 0 : Offline / Unknown]
     2. ControlFailure (Read only Status)
        a. [1 : Last control failed / Last control was Successful]
     3. ControlSelected (Read only Status)
        a. [1 : Select has been issued and execute pending , 0 : No select received yet]
     4. ControlExecute (External Control)
        a. [1 : User set to request execute , 0 : No Execute yet]
     5. ControlCancel (External Control)
        a. [1 : User set to Cancel last Select , 0 : No Cancel yet]
     6. Reserved
     7. Reserved
     8. Reserved
     9. Reserved
     10. Reserved
     11. Reserved
     12. Reserved
     13. Reserved
     14. Reserved
     15. Reserved
     16. Reserved

   - **Word 1 bit : [ADMIN.1]**
     For all below [1 : Request , 0 : Don't request]
     1. HardReset (External Request)
     2. Reconfigure (Internal Request) … not used at the moment
     3. RecoverAllAvailableData (Internal Request) … not used at the moment
     4. AcceptAndClearResetBit (Internal Request)
     5. TimeSync (Internal/External Request)
     6. StartUpPoll (Internal/External Request)
     7. PeriodicPoll (Internal/External Request)
     8. Class0DataPoll (Internal/External Request)
     9. Class1DataPoll (Internal/External Request)
     10. Class2DataPoll (Internal/External Request)

11. Class3DataPoll (Internal/External Request)
12. FailoverResynchronise(Internal/External Request)
13. EnableUnsolicited(Internal/External Request)
14. RequestLinkStatus((Internal/External Request)
15. ResetUSerProcess((Internal/External Request) not used.
16. DisableUnsolicted(Internal/External Request)

Note:
Status – For indication purposes and reporting to Adroit.

External – Bit will cause a specific control action within the driver such as a control or poll request.

Internal – Will most usually be set from within the driver itself due to protocol requirements such as IIN request bits.

Internal / External – Function may be safely called from Adroit but can also be called internally (driver) due to configuration and or IIN bit changes.

2. The driver supports the flag information that DNP provides as a separate data point with a "G" modifier on the address. e.g.

    30.2.2 U –> Analogue input
    30.2.2 UG -> Its corresponding flag. Note the G (separate scan point).

Each Flag is a bit packet set stored in an integer and is dependent on the corresponding object type, below is a list of each type (only the least significant 8 bits are used the rest are ignored and set to zero.)
(Marshals are recommended to get easy access to the bits).
Some objects are considered to be flagged as "normal/good" if they are sent in their bit packed form, e.g. Obj 1.1 and Obj 3.1. These points are considered online and normal on all other flag bits. If the flag is out of normal (say RemoteForcedData is true) this point will usually come in on Obj 1.2 to allow for this flag to be seen and used where applicable.

- **Analogue Outputs Flag** (Lsb to Msb)
    1. Online
    2. Restart
    3. ComLost
    4. RemoteForcedData
    5. Reserved
    6. Reserved
    7. Reserved
    8. Reserved

- **Analogue Inputs Flag** (Lsb to Msb)
    1. Online
    2. Restart
    3. ComLost
    4. RemoteForcedData
    5. LocalForcedData
    6. Overrange
    7. Reserved
    8. Reserved

- **Counter Inputs Flag** (Lsb to Msb)

1. Online
2. Restart
3. ComLost
4. RemoteForcedData
5. LocalForcedData
6. RollOver
7. Reserved
8. Reserved

- **Digital Flag** (Lsb to Msb)
    1. Online
    2. Restart
    3. ComLost
    4. RemoteForcedData
    5. LocalForcedData
    6. ChatterFilter
    7. Reserved
    8. State

- **Dual Bit Digital Flag** (Lsb to Msb)
    1. Online
    2. Restart
    3. ComLost
    4. RemoteForcedData
    5. LocalForcedData
    6. ChatterFilter
    7. State0
    8. State1

3. Select execute Control can be either internal or external depending on configuration (driver setup).
    a. If **internal** is selected, then the driver will verify via protocol and send the 2nd phase (Execute) immediately without user intervention. This will be virtually the same as a direct control action. Note the control bits in ADMIN.0 are therefore ignored, except the **controlFailed** bit that still reports success or failure.
    b. If **external** is selected, then when a select is detected (i.e. A control from Adroit on a specific tag is sent) the driver will issue a select to the device and upon receipt of a correct verification/echo will set the **[Selected bit – 3rd bit of ADMIN.0]**. This bit is available in Adroit (for example to be used for a visibility to enable a common execute button which will have an operator action that sets the **Execute bit**)
    c. Once the driver detects the **Execute bit** it will verify the selected bit is still on because the control could have timed out. The driver will then issue an execute to the device. Upon completion, fail or success, the select and execute will be cleared, and the **ControlFailed** bit will be set if the control failed, and it will stay set until the next successful control.

4. Supported Addresses

DNP data address types used as a reference for scanning by the Adroit DNP3 Protocol driver.

PLEASE NOTE

Other types are supported but for scanning purposes they will be stored in the tags with the address as described below. These can be viewed as "lowest common multiple", for example, event objects are stored in status objects, with the exception of frozen events which are considered separate datapoints and need their own tag and scan address.

Device		Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Digital					-			
Outputs	rw	10.2.0 A	10.2.x F	Digital	10.2.0 A	-	-	-
					-	-	-	-
Flags	r	10.2.x G	10.2.x G	Flag	-	10.2.0 G	-	-
Digital					-	-	-	-
Inputs	r	1.2.0	1.2.x	Digital	1.2.0	-	-	-
	r	1.2.0 G	1.2.x G	Flag	-	1.2.0 G	-	-
	r	3.2.0	3.2.x	Dual bit Digital	3.2.0	-	-	-
Flags	r	3.2.0 G	3.2.x G	Flag	-	3.2.0 G	-	-
Analogue								
Outputs	rw	40.2.0 U A	40.2.x U F	16bit	-	40.2.0 U A	40.2.0 U A	-
	rw	40.2.0 A	40.2.x F	16bit signed	-	40.2.0 A	40.2.0 A	-
	rw	40.1.0 U A	40.1.x U F	32bit	-	40.1.0 A	40.1.0 U A	-
	rw	40.1.0 A	40.1.x F	32bit signed	-	40.1.0 A	40.1.0 A	-
	rw	40.3.0 A	40.3.x F	32bit Float	-	40.3.0 A	40.3.0 A	-
	rw	40.4.0 A	40.4.x F	64bit Float	-	40.4.0 A	40.4.0 A	-
Flags	r	40.2.0 G	40.2.x G	Flag	-	40.2.0 G	40.2.0 G	-
	r	40.1.0 G	40.1.x G	Flag	-	40.1.0 G	40.1.0 G	-
Analogue								
Inputs static	r	30.2.0 U	30.2.x U	16bit	-	30.2.0 U	30.2.0 U	-
	r	30.2.0	30.2.x	16bit signed	-	30.2.0	30.2.0	-
	r	30.1.0 U	30.1.x U	32bit	-	30.1.0 U	30.1.0 U	-
	r	30.1.0	30.1.x	32bit signed	-	30.1.0	30.1.0	-
Flags	r	30.1.0 G	30.1.x G	Flag	-	30.1.0 G	30.1.0 G	-
	r	30.5.0	30.5.x	32bit Float	-	30.5.0	30.5.0	-
	r	30.6.0	30.6.x	64bit Float	-	30.6.0	30.6.0	-
Frozen	r	31.2.0 U	31.2.x U	16bit	-	31.2.0 U	31.2.0 U	-
	r	31.2.0	31.2.x	16bit signed	-	31.2.0	31.2.0	-
	r	31.1.0 U	31.1.x U	32bit	-	31.1.0 U	31.1.0 U	-
	r	31.1.0	31.1.x	32bit signed	-	31.1.0	31.1.0	-
Flags	r	31.1.0 G	31.1.x G	Flag	-	31.1.0 G	31.1.0 G	-
	r	31.7.0	30.7.x	32bit Float	-	30.7.0	30.7.0	-
	r	31.8.0	30.8.x	64bit Float	-	30.8.0	30.8.0	-
Counter								
Inputs	r	20.2.0 U	20.2.x U	16bit	-	20.2.0 U	20.2.0 U	-
	r	20.2.0	20.2.x	16bit signed	-	20.2.0	20.2.0	-
	r	20.1.0 U	20.1.x U	32bit	-	20.1.0 U	20.1.0 U	-
	r	20.1.0	20.1.x	32bit signed	-	20.1.0	20.1.0	-
Flags	r	20.1.0 G	20.1.x G	Flag	-	20.1.0 G	20.1.0 G	-
Frozen	r	21.2.0 U	21.2.x U	16bit	-	21.2.0 U	21.2.0 U	-
	r	21.2.0	21.2.x	16bit signed	-	21.2.0	21.2.0	-
	r	21.1.0 U	21.1.x U	32bit	-	21.1.0 U	21.1.0 U	-
	r	21.1.0	21.1.x	32bit signed	-	21.1.0	21.1.0	-

Flags	r	21.1.0 G	21.1.x G	Flag	-	21.1.0 G	21.1.0 G	-
Psuedo								
Control Words	rw	ADMIN.0	ADMIN.1	16Bit Marshals	-	ADMIN.0	-	-

X: If Index is 8bit it is 255 16 bit it is 65535.
r: Read Only.
rw: Read and Writeable.

4.1. Advanced Registry Key Settings

MarkBadTolerance (DWORD)

HKEY_LOCAL_MACHINE\SOFTWARE\Adroit Technologies\Adroit\ProtocolDrivers\DNP3\<devicename>\Primary\ MarkBadTolerance

(Default = 1) No change mark device bad on any single failure.

If set to 0, disable bad marking completely.

WARNING

This will result in the Device Agent's communications failed bit to never be set during a communication fail. Admin word 0, bit 0 is still updated if a communications failure is detected, and this bit shall be used to determine online status.

Any value greater than 1, the number of final failures (only final retries are counted and only consecutive fails are counted) will determine the mark bad status.

Example:
MarkBadTolerance = 3, Retries = 3;

The device communication must fail the next consecutive transactions. 3 messages and 3 retries means (3 × (3+1) = 12). If the count is at 10 and the next message response is in protocol then the count is reset and the markbad will **NOT** take place.

In summary, the device will only be marked bad after on the 13th consecutive failed message.

5. Protocol Definition and Considerations

See DNP3 Users group documentation.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)

7. Appendix A – Master Profile Document

DNP V3.00 **MASTER PROFILE DOCUMENT** This document must be accompanied by a table having the following headings: See Appendix B	
Highest DNP Level Supported: For Requests Level 2 + (some 3 and 4) For Responses Level 2 + (some 3 and 4)	Device Function: ☑ Master ☐ Slave
Notable objects, functions, and/or qualifiers supported in addition to the Highest DNP Levels Supported (the complete list is described in the attached table): Additional variations on 30,31,32,33,40 that refer to 32bit/64bit Floating point numbers. The ALTERNATE floating point object are NOT supported as yet. (100,101) The Driver can parse any valid qualifier code (more) than specified in level2, as well as more variations than specified (referred to above). Frozen points are not considered the same point as their static counterparts , these will be in separate tags. Any Flag information is also available as required , but will be stored in a separate tag (INTEGER) see prefix G.	
Maximum Data Link Frame Size (octets): Transmitted: 292 Received: 292 (must be 292)	Maximum Application Fragment Size (octets): Transmitted : Only ever needs 1 transport fragment. Received :Unlimited /driven by protocol sequencing.
Maximum Data Link Re-tries: ☑ None ☐ Fixed at ________________________ ☐ Configurable, range ____ to _______	Maximum Application Layer Re-tries: ☐ None ☑ Configurable, range 0 to MAX_WORD (Fixed is not permitted)
Requires Data Link Layer Confirmation: ☐ Never ☐ Always ☑ Sometimes **It is better to disable link requests.**	

The master sends a link reset at startup **this is fixed at this time** but will be configurable in the next revision.
The master will provide Link Layer Confirmation when/if the slave requests it.
The master also now send link acks to link status requests from the slave units.

☐Configurable If 'Configurable', how? ____

Requires Application Layer Confirmation:

☐ Never
☑ Always
☐ When reporting Event Data (Slave devices only)
☐ When sending multi-fragment responses (Slave devices only)

☐ Sometimes

☐ Configurable   If 'Configurable', how? ____

Timeouts while waiting for:

Data Link Confirm	☐ None	☐ Fixed at _________	☐ Variable	☑ Configurable
Complete Appl. Fragment	☑ None	☐ Fixed at _________	☐ Variable	☐ Configurable
Application Confirm	☐ None	☐ Fixed at _________	☐ Variable	☑ Configurable
Complete Appl. Response	☐ None	☐ Fixed at _________	☐ Variable	☑ Configurable

Others : The Complete Application response time is the total time to wait for ALL application fragments, to complete a request.(However each application fragment is parsed separately upon receipt of the final transport fragment).
Data Link Confirm and Application confirm is the same timeout configuration value.

Attach explanation if 'Variable' or 'Configurable' was checked for any timeout

Sends/Executes Control Operations:

WRITE Binary Outputs	☐ Never	☐ Always	☐ Sometimes	☑ Configurable
SELECT/OPERATE	☐ Never	☐ Always	☐ Sometimes	☑ Configurable
DIRECT OPERATE	☐ Never	☐ Always	☐ Sometimes	☑ Configurable
DIRECT OPERATE - NO ACK	☐ Never	☐ Always	☐ Sometimes	☑ Configurable

Count > 1	☑ Never	☐ Always	☐ Sometimes	☐ Configurable
Pulse On	☐ Never	☐ Always	☐ Sometimes	☑ Configurable
Pulse Off	☐ Never	☐ Always	☐ Sometimes	☑ Configurable
Latch On	☐ Never	☐ Always	☐ Sometimes	☑ Configurable
Latch Off	☐ Never	☐ Always	☐ Sometimes	☑ Configurable

Queue	☑ Never	☐ Always	☐ Sometimes	☐ Configurable
Clear Queue	☑ Never	☐ Always	☐ Sometimes	☐ Configurable

Attach explanation if 'Sometimes' or 'Configurable' was checked for any operation. Each Control tag has a postfix (“A” – “F”) that represents up to 6 control types individually configurable e.g. A - Direct operate - Pulse - on Time - off Time.
FILL OUT THE FOLLOWING ITEM FOR MASTER DEVICES ONLY:
Expects Binary Input Change Events: ☐ Either time-tagged or non-time-tagged for a single event ☑ Both time-tagged and non-time-tagged for a single event ☐ Configurable (attach explanation) Tags are defined and linked to the most basic type, if events come in with time these are treated as the same point and stored accordingly. **Events that do not have time are treated on a time of arrival basis.**

8. Appendix B - DNP Subset Definitions

OBJECT			REQUEST (slave must parse)		RESPONSE (master parses)	
Obj	**Var**	**Description**	**Function Codes (dec)**	**Qual Codes (hex)**	**Function Codes**	**Qual Codes (hex)**
1	0	Binary Input - All Variations	1	06		
1	1	Binary Input			129, 130	00, 01
1	2	Binary Input with Status			129, 130	00, 01
2	0	Binary Input Change - All Variations	1	06,07,08		
2	1	Binary Input Change without Time	1	06,07,08	129, 130	17, 28,*
2	2	Binary Input Change with Time	1	06,07,08	129, 130	17, 28,*
2	3	Binary Input Change with Relative Time	1	06,07,08	129, 130	17, 28
3	0	Double bit Binary Input - All Variations	1	06		
3	1	Double bit Binary Input			129, 130	00, 01
3	2	Double bit Binary Input with Status			129, 130	00, 01
4	0	Double bit Binary Input Change - All Variations	1	06,07,08		
4	1	Double bit Binary Input Change without Time	1	06,07,08	129, 130	17, 28,*
4	2	Double bit Binary Input Change with Time	1	06,07,08	129, 130	17, 28,*
4	3	Double bit Binary Input Change with Relative Time	1	06,07,08	129, 130	17, 28
10	0	Binary Output - All Variations	1	06		
10	1	Binary Output				
10	2	Binary Output Status	1	00,01,06	129, 130	00, 01
12	0	Control Block - All Variations				
12	1	Control Relay Output Block	3, 4, 5, 6	17, 28	129	echo of request
12	2	Pattern Control Block				
12	3	Pattern Mask				
20	0	Binary Counter - All Variations	1, 7, 8, 9, 10	06		
20	1	32-Bit Binary Counter			129, 130	00, 01
20	2	16-Bit Binary Counter			129, 130	00, 01
20	3	32-Bit Delta Counter			129, 130	00, 01
20	4	16-Bit Binary Counter			129, 130	00, 01
20	5	32-Bit Binary Counter without Flag			129, 130	00, 01
20	6	16-Bit Binary Counter without Flag			129, 130	00, 01
20	7	32-Bit Delta Counter without Flag			129, 130	00 ,01
20	8	16-Bit Delta Counter without Flag			129, 130	00 ,01
21	0	Frozen Counter - All Variations	1	06		
21	1	32-Bit Frozen Counter			129, 130	00, 01
21	2	16-Bit Frozen Counter			129, 130	00, 01
21	3	32-Bit Frozen Delta Counter			129, 130	00, 01
21	4	16-Bit Frozen Delta Counter			129, 130	00, 01

OBJECT			REQUEST (slave must parse)		RESPONSE (master parses)	
Obj	**Var**	**Description**	**Function Codes (dec)**	**Qual Codes (hex)**	**Function Codes**	**Qual Codes (hex)**
21	5	32-Bit Frozen Counter with Time of Freeze				
21	6	16-Bit Frozen Counter with Time of Freeze				
21	7	32-Bit Frozen Delta Counter with Time of Freeze				
21	8	16-Bit Frozen Delta Counter with Time of Freeze				
21	9	32-Bit Frozen Counter without Flag			129, 130	00, 01
21	10	16-Bit Frozen Counter without Flag			129, 130	00, 01
21	11	32-Bit Frozen Delta Counter without Flag				
21	12	16-Bit Frozen Delta Counter without Flag				
22	0	Counter Change Event - All Variations	1	06,07,08		
22	1	32-Bit Counter Change Event without Time			129, 130	17, 28
22	2	16-Bit Counter Change Event without Time			129, 130	17, 28
22	3	32-Bit Delta Counter Change Event without Time			129, 130	17, 28
22	4	16-Bit Delta Counter Change Event without Time			129, 130	17, 28
22	5	32-Bit Counter Change Event with Time			129, 130	17, 28
22	6	16-Bit Counter Change Event with Time			129, 130	17, 28
22	7	32-Bit Delta Counter Change Event with Time			129, 130	17, 28
22	8	16-Bit Delta Counter Change Event with Time			129, 130	17, 28
23	0	Frozen Counter Event - All Variations	1	06,07,08		
23	1	32-Bit Frozen Counter Event without Time			129, 130	17, 28
23	2	16-Bit Frozen Counter Event without Time			129, 130	17, 28
23	3	32-Bit Frozen Delta Counter Event without Time			129, 130	17, 28
23	4	16-Bit Frozen Delta Counter Event without Time			129, 130	17, 28
23	5	32-Bit Frozen Counter Event with Time			129, 130	17, 28
23	6	16-Bit Frozen Counter Event with Time			129, 130	17, 28
23	7	32-Bit Frozen Delta Counter Event with Time			129, 130	17, 28
23	8	16-Bit Frozen Delta Counter Event with Time			129, 130	17, 28
30	0	Analog Input - All Variations	1	06		
30	1	32-Bit Analog Input			129, 130	00, 01
30	2	16-Bit Analog Input			129, 130	00, 01
30	3	32-Bit Analog Input without Flag			129, 130	00, 01
30	4	16-Bit Analog Input without Flag			129, 130	00, 01
30	5	32-Bit Single precision Float Input with Flag			129, 130	00, 01
30	6	64-Bit Double precision Float Input with Flag			129, 130	00, 01
31	0	Frozen Analog Input - All Variations	1	06		
31	1	32-Bit Frozen Analog Input			129, 130	00, 01
31	2	16-Bit Frozen Analog Input			129, 130	00, 01
31	3	32-Bit Frozen Analog Input with Time of Freeze			129, 130	00, 01
31	4	16-Bit Frozen Analog Input with Time of Freeze			129, 130	00, 01

OBJECT			REQUEST (slave must parse)		RESPONSE (master parses)	
Obj	**Var**	**Description**	**Function Codes (dec)**	**Qual Codes (hex)**	**Function Codes**	**Qual Codes (hex)**
31	5	32-Bit Frozen Analog Input without Flag			129, 130	00, 01
31	6	16-Bit Frozen Analog Input without Flag			129, 130	00, 01
31	7	32-Bit Frozen Analog Input Float without Flag			129, 130	00, 01
31	8	64-Bit Frozen Analog Input Float without Flag			129, 130	00, 01
32	0	Analog Change Event - All Variations	1	06,07,08		
32	1	32-Bit Analog Change Event without Time			129,130	17,28
32	2	16-Bit Analog Change Event without Time			129,130	17,28
32	3	32-Bit Analog Change Event with Time			129,130	17,28
32	4	16-Bit Analog Change Event with Time			129,130	17,28
32	5	2-Bit Single precision float Change Event with Flag			129,130	17,28
32	6	64-Bit Double precision float Change Event with Flag			129,130	17,28
32	7	32-Bit Single precision float Change Event with Flag and Time			129,130	17,28
32	8	64-Bit Double precision float Change Event with Flag and Time			129,130	17,28
33	0	Frozen Analog Event - All Variations	1	06,07,08		
33	1	32-Bit Frozen Analog Event without Time			129,130	17,28
33	2	16-Bit Frozen Analog Event without Time			129,130	17,28
33	3	32-Bit Frozen Analog Event with Time			129,130	17,28
33	4	16-Bit Frozen Analog Event with Time			129,130	17,28
33	5	32-Bit Frozen Analog Change Event with Flag			129,130	17,28
33	6	64-Bit Frozen Analog Change Event with Flag			129,130	17,28
33	7	32-Bit Frozen Analog Change Event with Flag and Time			129,130	17,28
33	8	64-Bit Frozen Analog Change Event with Flag and Time			129,130	17,28
40	0	Analog Output Status - All Variations	1	06		
40	1	32-Bit Analog Output Status			129, 130	00, 01
40	2	16-Bit Analog Output Status			129, 130	00, 01
40	3	32-Bit Analog Output Float Status			129, 130	00, 01
40	4	64-Bit Analog Output Float Status			129, 130	00, 01
41	0	Analog Output Block - All Variations				
41	1	32-Bit Analog Output Block	3, 4, 5, 6	17, 28	129	echo of request
41	2	16-Bit Analog Output Block	3, 4, 5, 6	17, 28	129	echo of request
50	0	Time and Date - All Variations				
50	1	Time and Date	2	07 where quantity = 1		
50	2	Time and Date with Interval				
51	0	Time and Date CTO - All Variations				
51	1	Time and Date CTO			129, 130	07, quantity=1

OBJECT			REQUEST (slave must parse)		RESPONSE (master parses)	
Obj	**Var**	**Description**	**Function Codes (dec)**	**Qual Codes (hex)**	**Function Codes**	**Qual Codes (hex)**
51	2	Unsynchronized Time and Date CTO			129, 130	07, quantity=1
52	0	Time Delay - All Variations				
52	1	Time Delay Coarse			129	07, quantity=1
52	2	Time Delay Fine			129	07, quantity=1
60	0					
60	1	Class 0 Data	1	06		
60	2	Class 1 Data	1	06,07,08		
60	3	Class 2 Data	1	06,07,08		
60	4	Class 3 Data	1	06,07,08		
70	1	File Identifier				
80	1	Internal Indications	2	00 index=7		
81	1	Storage Object				
82	1	Device Profile				
83	1	Private Registration Object				
83	2	Private Registration Object Descriptor				
90	1	Application Identifier				
100	1	Short Floating Point				
100	2	Long Floating Point				
100	3	Extended Floating Point				
101	1	Small Packed Binary-Coded Decimal				
101	2	Medium Packed Binary-Coded Decimal				
101	3	Large Packed Binary-Coded Decimal				
		No Object	13			
		No Object	23			

9. Revisions

9.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.2	19-Jan-2023	K. Robinson	Grammar edits, content updates and improvements.	D. Jordaan
2.0				

9.2. DLL Revision

Rev.	Date	Changes
91	26-July 2021	Schemas when using B through F and especially multiple schemas on one device agent would not be stored correctly. Schema A would normally be unaffected. Fixed. Multiple Tags scanned to the same DNP address will override the schema for all duplicated tags, the schema of last tag to be scanned rules. The application event log will have a warning entry describing the duplication, as will the Datascope output.
96	05-August 2022	Added separation capability for Class0 and Integrity paradigms see advanced dialogue. Thread clean up fixed. Enable override of quality flag if object doesn't set it. Application sequencing failed on keepalive link messaging. Added cluster support.
97	10-Oct 2022	Added hidden registry UnsolicitedIgnoresMarkBadTolerance opt in